

Newman-Penrose Formalism Calculations

Stephani et al., Equation (37.55_1)

Metric

$$ds^2 = (-br^2 - a) dt \otimes dt + \left(\frac{2br^2 + a}{br^2 + a} \right) dr \otimes dr + \left(-\frac{r^2}{\rho^2 - 1} \right) d\rho \otimes d\rho + r^2 \rho^2 d\phi \otimes d\phi$$

Coordinates

$$(x^\mu) = (t, r, \rho, \phi)$$

Null Tetrad

Covariant components

$$l = \left(\frac{1}{2} \sqrt{2} \sqrt{br^2 + a} \right) dt + \left(\frac{\sqrt{2} \sqrt{2br^2 + a}}{2 \sqrt{br^2 + a}} \right) dr$$

$$n = \left(\frac{1}{2} \sqrt{2} \sqrt{br^2 + a} \right) dt + \left(-\frac{\sqrt{2} \sqrt{2br^2 + a}}{2 \sqrt{br^2 + a}} \right) dr$$

$$m = \left(\frac{\sqrt{2}r}{2 \sqrt{-\rho^2 + 1}} \right) d\rho + \left(\frac{1}{2} i \sqrt{2} r \rho \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}r}{2 \sqrt{-\rho^2 + 1}} \right) d\rho + \left(-\frac{1}{2} i \sqrt{2} r \rho \right) d\phi$$

Contravariant components

$$l = \left(-\frac{\sqrt{2}}{2\sqrt{br^2+a}} \right) dt + \left(\frac{\sqrt{2}\sqrt{br^2+a}}{2\sqrt{2}br^2+a} \right) dr$$

$$n = \left(-\frac{\sqrt{2}}{2\sqrt{br^2+a}} \right) dt + \left(-\frac{\sqrt{2}\sqrt{br^2+a}}{2\sqrt{2}br^2+a} \right) dr$$

$$m = \left(\frac{\sqrt{2}\sqrt{\rho+1}\sqrt{-\rho+1}}{2r} \right) d\rho + \left(\frac{i\sqrt{2}}{2r\rho} \right) d\phi$$

$$\bar{m} = \left(\frac{\sqrt{2}\sqrt{\rho+1}\sqrt{-\rho+1}}{2r} \right) d\rho + \left(-\frac{i\sqrt{2}}{2r\rho} \right) d\phi$$

Directional Derivatives

$$D = -\frac{\sqrt{2}}{2\sqrt{br^2+a}} \partial_t + \frac{\sqrt{2}\sqrt{br^2+a}}{2\sqrt{2}br^2+a} \partial_r$$

$$\Delta = -\frac{\sqrt{2}}{2\sqrt{br^2+a}} \partial_t - \frac{\sqrt{2}\sqrt{br^2+a}}{2\sqrt{2}br^2+a} \partial_r$$

$$\delta = \frac{\sqrt{2}\sqrt{\rho+1}\sqrt{-\rho+1}}{2r} \partial_\rho + \frac{i\sqrt{2}}{2r\rho} \partial_\phi$$

$$\bar{\delta} = \frac{\sqrt{2}\sqrt{\rho+1}\sqrt{-\rho+1}}{2r} \partial_\rho - \frac{i\sqrt{2}}{2r\rho} \partial_\phi$$

Spin Coefficients

$$\rho = -\frac{\sqrt{2}\sqrt{br^2+a}}{2\sqrt{2}br^2+ar}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = -\frac{\sqrt{2}\sqrt{br^2+a}}{2\sqrt{2br^2+ar}}$$

$$\lambda = 0$$

$$\nu = 0$$

$$\alpha = \frac{\sqrt{2}\sqrt{\rho+1}(\rho-1)}{4r\rho\sqrt{-\rho+1}}$$

$$\beta = -\frac{\sqrt{2}\sqrt{\rho+1}(\rho-1)}{4r\rho\sqrt{-\rho+1}}$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = \frac{\sqrt{2}br}{4\sqrt{2br^2+a}\sqrt{br^2+a}}$$

$$\gamma = \frac{\sqrt{2}br}{4\sqrt{2br^2+a}\sqrt{br^2+a}}$$

Ricci Tensor Components

$$\Phi_{00} = \frac{(br^2 + a)b}{(2br^2 + a)^2}$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{(br^2 + a)b}{2(2br^2 + a)^2}$$

$$\Phi_{12} = 0$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = 0$$

$$\Phi_{22} = \frac{(br^2 + a)b}{(2br^2 + a)^2}$$

$$\Lambda = -\frac{b^2r^2}{6(2br^2 + a)^2}$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = -\frac{2\,b^2r^2}{3\,(2\,br^2+a)^2}$$

$$\Psi_3 = 0$$

$$\Psi_4 = 0$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "D"